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A Search for GeV Emission from Magnetar Giant Flare Candidates with Fermi-LAT

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Abstract

The discovery of delayed GeV emission from the extragalactic magnetar giant flare (MGF) GRB 200415A, located in the nearby Sculptor galaxy, revealed for the first time that these rare transients can launch relativistic outflows that power high-energy afterglows. Motivated by the recent identification of additional nearby MGF candidates in the archival data of the Fermi Gamma-ray Burst Monitor, we conduct a search for GeV counterparts with the Fermi Large Area Telescope (LAT). We analyzed post-trigger time intervals taken in the range 10^2–10^4 s using a maximum-likelihood approach and performed a stacking analysis of all candidates with LAT coverage. In addition, we searched for photon triplets through a waiting time analysis to identify events potentially associated with MGFs. We recover the known delayed signal from GRB 200415A but find no GeV emission from the remaining six candidates. For three events, the earliest emission is unconstrained because the 10^2 s interval contains zero exposure after standard selections. For the events with LAT coverage, we obtain upper limits at 95% confidence level of order $F_E \sim 10^{-9}$ erg cm^{-2} s^{-1} for the individual events, and a stacked population-averaged limit of $\approx 2 \times 10^{-10}$ erg cm^{-2} s^{-1}. We interpret these upper limits within the relativistic fireball framework, where the prompt spectral peaks favor a baryonic-poor regime ($\eta > \eta_*$). For the candidates with hard prompt spectrum and early LAT coverage, the baryonic-poor condition restricts the mass of relativistic ejecta to $M_b \lesssim 1$–4×10^{22} g; the LAT upper limits confirm that the predicted GeV afterglow from such clean outflows falls below current instrumental sensitivity.

Unified Astronomy Thesaurus concepts: Magnetars (992); Gamma-ray bursts (629); Gamma-ray astronomy (628); High energy astrophysics (739); Transient sources (1851); Relativistic jets (1390)

1. Introduction

Magnetars are neutron stars that exhibit both transient soft gamma-ray activity and persistent X-ray emission. Their emission is driven by the evolution and decay of ultrastrong magnetic fields ($B \gtrsim 5 \times 10^{13}$–$10^{15}$G) (see, e.g., V. M. Kaspi & A. M. Beloborodov 2017 and M. Negro et al. 2024 for a review). While they typically exhibit persistent X-ray emission ($L_X \sim 10^{33}$–10^{36} erg s^{-1}), long-period brightening (outbursts), and short bursts, their most extreme manifestations are magnetar giant flares (MGFs). These events release energies of the order of 10^{44}–10^{47} erg within a fraction of a second, characterized by a hard, nonthermal initial spike followed by a softer, pulsating tail modulated at the stellar spin period (E. P. Mazets et al. 1979; K. Hurley et al. 1999; D. M. Palmer et al. 2005).

The interaction of MGF ejecta with the surrounding medium is expected to drive shocks capable of particle acceleration. Radio observations of the 2004 giant flare from SGR 1806−20 revealed an expanding nebula consistent with a mildly relativistic outflow ($\Gamma \sim 1.5$) and significant baryon loading (B. M. Gaensler et al. 2005; J. D. Gelfand et al. 2005). However, theoretical models suggest that MGFs may also launch ultrarelativistic outflows ($\Gamma \gg 10$) analogous to those in short gamma-ray bursts (GRBs), which would generate high-energy (> 100 MeV) emission via synchrotron or inverse-Compton processes at the external shock (J. Granot et al. 2006; M. Lyutikov 2006).

The identification of GRB 200415A, spatially coincident with the galaxy NGC 253 ($D \approx 3.7$ Mpc), as an extragalactic MGF established a new observational class (J. Yang et al. 2020; D. Svinkin et al. 2021). Crucially, subsequent Fermi Large Area Telescope (LAT) analysis revealed delayed ($\sim$19–380 s) GeV emission with a hard spectrum ($\Gamma_{\rm LAT} \simeq -1.7$), consistent with synchrotron radiation from an external shock or a bow shock driven by relativistic ejecta (H.-M. Zhang et al. 2020; V. Chand et al. 2021; Fermi-LAT Collaboration et al. 2021). This discovery provided the first confirmed GeV counterpart to an MGF and established a framework for similar searches.

This GeV emission can offer a direct probe to the baryonic content in the outflow. In the "fireball" model, the presence of a prompt spike with a quasi-thermal spectrum requires a "baryonic-poor" outflow (high dimensionless entropy η) that follows the $E_{\rm p} \propto E_{\rm iso}^{1/4}$ relation. On the other hand, the generation of a bright external shock afterglow requires sufficient kinetic energy, which scales with the baryonic mass (K. Ioka et al. 2005; E. Nakar et al. 2005). Observational constraints on the GeV flux thus define the allowable phase space for the ejecta mass and Lorentz factor, distinguishing between baryonic-rich and baryonic-poor regimes (H.-M. Zhang et al. 2020).

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Table 1
Nearby MGF Candidates: GBM Triggers, Host Galaxies, and Prompt Emission Properties

GRB	GBM Trigger ID	Host Galaxy	R.A.$_{\rm gal}$ (deg)	Decl.$_{\rm gal}$ (deg)	d (Mpc)	T_{90} (s)	$E_{\rm p}$ (keV)	Index$^{\alpha}$	$E_{\gamma,45}$ (10^{45} erg)	References
GRB 081213A	bn081213173	NGC 253	11.8881	−25.2888	3.7	0.050	400	−0.6	0.42	1
GRB 180128A	bn180128215	NGC 253	11.8881	−25.2888	3.7	0.155	290	−0.6	0.60	2
GRB 200415A	bn200415367	NGC 253	11.8881	−25.2888	3.7	0.200	1080	−0.0	14.2	3
GRB 231024A	bn231024556	NGC 253	11.8881	−25.2888	3.7	0.094	500	−0.8	0.55	4
GRB 120616A	bn120616630	IC 342	56.7021	+ 68.0961	2.3	0.050	500	−0.4	0.23	4
GRB 200423A	bn200423579	NGC 6946	308.7181	+ 60.1537	7.7	0.032	600	−0.7	8.50	4
GRB 231115A	bn231115650	M82	148.9685	+ 69.6797	3.5	0.097	600	−0.1	1.15	5

Note. $E_{\gamma,45} \equiv E_{\gamma,\rm iso}/10^{45}$ erg. Distances are approximate values for the host galaxies.
References. (1) E. Bissaldi & A. von Kienlin (2008); (2) A. C. Trigg et al. (2024); (3) Fermi-LAT Collaboration et al. (2021); (4) A. C. Trigg et al. (2026); (5) A. C. Trigg et al. (2025).

Recently, A. C. Trigg et al. (2026) systematically reanalyzed the entire Fermi Gamma-ray Burst Monitor (GBM) (C. Meegan et al. 2009) short-GRB catalog to identify previously overlooked extragalactic MGF candidates. They uncovered four new events, GRBs 081213A, 120616A, 200423A, and 231024A, consistent with the known MGF population in temporal and spectral properties. Together with the three previously identified events (GRB 200415A, GRB 180128A, and GRB 231115A), these findings expand the total extragalactic MGF sample to 10 events, which includes three identifications from before the launch of Fermi. This newly identified sample of nearby extragalactic MGFs motivates dedicated high-energy searches in the Fermi-LAT (W. B. Atwood et al. 2009) data to probe whether delayed GeV emission, like that observed from GRB 200415A, is a common feature of these flares.

In this paper, we present a systematic search for GeV emission from this expanded sample using Fermi-LAT data. We perform unbinned likelihood analyses and stacked-likelihood, photon-triplet analyses to probe both individual and cumulative signals to constrain the high-energy properties of these flares. We outline our data analysis in Section 2, present the search results in Section 3, discuss the implications for the baryon loading and energetics of MGF outflows in Section 4, and draw our conclusions in Section 5.

2. Data and Methods

We analyze Fermi-LAT data for the seven GBM triggers listed in Table 1, focusing on three post-trigger integration windows, $\Delta T = \{10^2, 10^3, 10^4\}$ s relative to the GBM trigger time. Data were processed with the `gtburst` GUI (G. Vianello 2016) distributed with the `Fermitools` software package (version 2.2.0; Fermi Science Support Development Team 2019), following standard LAT GRB methodology (M. Ajello et al. 2019). We use Pass 8 (P8R3) processed data selecting transient event classes and the corresponding instrument response functions (IRFs) (W. Atwood et al. 2013; P. Bruel et al. 2018). In particular, we use the `P8R3_TRANSIENT020E_V3` event class for the shortest window and the more stringent `P8R3_TRANSIENT010E_V3` event class for the longer windows. In all selections we utilize the FRONT+BACK event type (`evtype=3`). The maximum likelihood analysis is performed over 0.1–100 GeV, while we report photon/energy fluxes integrated over 0.1–10 GeV as described in Section 2.1; the highest-energy photon associated with GRB 200415A was ∼1.7 GeV (Fermi-LAT Collaboration et al. 2021), and no events above 10 GeV are detected for any candidate, so the 0.1–10 GeV sub-band is conservatively used to report fluxes. We adopt a region of interest (ROI) of radius 12° centered on the MGF localization (host-galaxy coordinates). To suppress contamination from the Earth limb, a maximum zenith angle of $z_{\rm max} = 100°$ is applied. The sky model includes the Galactic diffuse emission template `gll_iem_v07.fits` and the isotropic component matching the event selection, together with all 4FGL-DR4 catalog sources within the ROI (S. Abdollahi et al. 2022; J. Ballet et al. 2023). Normalizations of sources within 5° of the ROI center are left free in the fit, while parameters of more distant sources are frozen to their catalog values. Normalizations of the diffuse components are left free unless otherwise noted. Throughout this work, upper limits (ULs) are computed with the photon index fixed to $\Gamma = -2$, broadly consistent with the measured power-law index of GRB 200415A ($\Gamma \simeq -1.7 \pm 0.3$; Fermi-LAT Collaboration et al. 2021). This convention applies to individual limits (Section 2.1), stacked limits, and bin-by-bin spectral energy distribution (SED) scans (Section 2.2). Because the LAT field of view does not continuously cover a given sky position, some events have delayed or partial exposure within the nominal integration windows. In such cases, the likelihood analysis uses only the available good time intervals (GTIs), and the reported ULs apply to those observed epochs; emission prior to the first exposure remains unconstrained.

2.1. Detection and Upper Limits

For each MGF candidate and integration window, we perform unbinned likelihood fits using `gtlike`, adopting a `PowerLaw2` spectral model that parameterizes the integrated photon flux between 0.1 and 100 GeV. The detection significance is quantified with the test statistic (TS), defined as $\mathrm{TS} = 2\Delta \ln \mathcal{L}$ between models with and without the source (e.g., J. R. Mattox et al. 1996). For each ROI, we generate TS maps both with and without the inclusion of the candidate source in the model; note that some host-galaxy positions coincide with known 4FGL sources. For these TS maps, the normalization and spectral parameters of all 4FGL sources are frozen to their catalog values, and the diffuse components are fixed to the best-fit values from the corresponding likelihood fit.

For nondetections, we derive 95% confidence level (CL) ULs on the 0.1–10 GeV integrated flux via profile likelihood (Figure 1), scanning the source flux while refitting the diffuse components and taking the flux value where $\Delta\mathrm{TS} = 2.71$

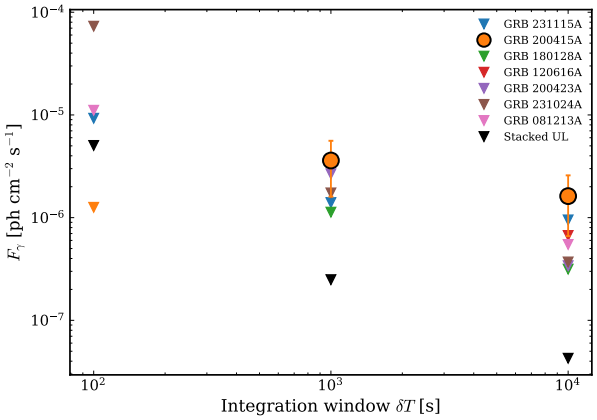


Figure 1. Photon fluxes (0.1–10 GeV) from unbinned likelihood fits in post-trigger windows $\Delta T = 10^2$, 10^3, 10^4 s. Filled circles show the GRB 200415A detections (1σ statistical uncertainties). Colored inverted triangles show 95% confidence ULs for the individual nondetected candidates (photon index fixed to $\Gamma = -2$); symbols with black outlines could be included in the stacked-likelihood analysis for the corresponding time window. Black inverted triangles show the 95% confidence ULs from the stacked analysis. Note that at $\Delta T = 10^2$ s only two events (GRBs 081213A and 231115A) effectively contribute to the stacking likelihood; the remaining candidate (GRB 231024A) has zero detected counts and nonzero exposure, so its individual UL—derived from Poisson statistics on the exposure alone—is inherently loose and does not meaningfully tighten the stacked constraint.

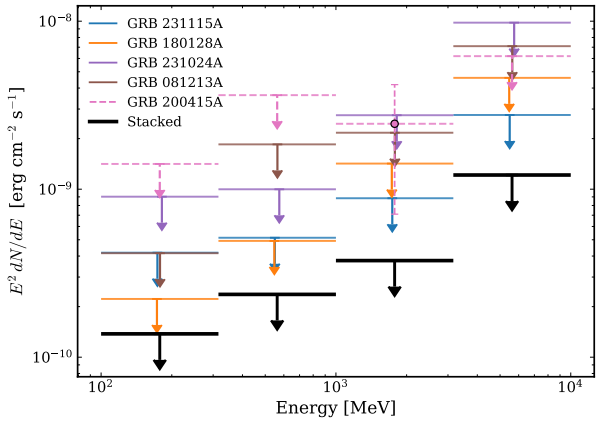


Figure 2. SED constraints in a 500 s post-trigger interval. Downward arrows indicate 95% CL ULs in each energy bin from the profile-likelihood scans with the index fixed to $\Gamma = -2$. The stacked limits combine the four nondetected events included in the 500 s stack; GRB 200415A is shown for reference but excluded from the stacking.

(corresponding to the 95% CL for one parameter of interest). In addition, for each burst we construct an SED for the first 500 s after trigger by dividing the LAT energy range into two logarithmic bins per decade. In each bin we refit the normalization with the index fixed to the global best-fit value (for GRB 200415A) or to the fiducial value (for nondetections), and compute ULs via the same profile-likelihood method.

2.2. Stacking Analysis

To search for a cumulative GeV signal from the MGF candidate population, we perform a stacked-likelihood analysis following the approach commonly used in LAT population studies (e.g., V. S. Paliya et al. 2019; G. Principe et al. 2023). For each integration window, the combined likelihood is taken as the product of the individual likelihoods for the subset of bursts with LAT exposure,

$$\mathcal{L}_{\rm stack}(F) = \prod_k \mathcal{L}_k(F), \tag{1}$$

and the stacked test statistic is defined as

$$\mathrm{TS}_{\rm stack} = 2[\ln \mathcal{L}_{\rm stack}(F) - \ln \mathcal{L}_{\rm stack}(0)]. \tag{2}$$

Here, F is the common (population-averaged) photon flux, and each $\mathcal{L}_k$ is evaluated by scanning F while refitting nuisance parameters in the background model (diffuse normalizations and relevant nearby sources), with the photon index fixed to -2.

In the absence of a significant collective detection (we adopt $\mathrm{TS}_{\rm stack} < 25$ as a nondetection criterion), we derive the 95% CL UL $\hat{F}_{95}$ on the common flux from the stacked profile likelihood by using the same profile-likelihood method described in Section 2.1. We also produce stacked SEDs over 500 s intervals by performing the same likelihood scan independently in each energy bin across all stacked bursts (bin-by-bin SED). For the 500 s integration, we can stack four events (Figure 2); GRB 200415A is excluded from the stacking because it is a known LAT detection and would dominate the combined constraint, whereas the remaining candidates yield only ULs in the GeV band. The two candidates excluded from the 500 s stack (GRBs 120616A and 200423A) lack LAT coverage within the first 500 s post-trigger (Table 2).

2.3. Photon-triplet Counting Analysis

We perform a model-independent photon-triplet search designed to identify short-duration GeV emission that may be temporally coincident with MGF events. This approach complements the time-integrated likelihood analysis by providing sensitivity to brief, intense GeV flares that could be diluted in longer integration windows. The method was originally developed for the Fermi-LAT analysis of GRB 200415A (Fermi-LAT Collaboration et al. 2021) and subsequently adapted for searches for gamma-ray counterparts to fast radio bursts (G. Principe et al. 2023). The procedure is the following.

For each MGF candidate, we extract all SOURCE-class (`P8R3_SOURCE`) LAT photons in the energy range 0.1–100 GeV within a circular ROI of radius 1° centered on the host-galaxy coordinates. We search for sets of three consecutive photons (hereafter "triplets") in the selected region around the target.

For a given set of three consecutive photons indexed as i, $i + 1$, and $i + 2$, we define the triplet time interval as

$$\Delta t_i = t_{i+2} - t_i, \tag{3}$$

which characterizes the waiting time of the triplet. Under the assumption of a steady Poisson background with mean rate R (estimated from the ROI and observation period), Δt is the sum of two exponential waiting times and is therefore Erlang-2 distributed, with probability density

$$P(\Delta t) = R^2\ \Delta t\ e^{-R\Delta t}, \tag{4}$$

Table 2
Fermi-LAT Analysis Results

GRB	Time Window (s)	TS	Photon Flux (10^{-6} ph cm^{-2} s^{-1})	Energy Flux (10^{-9} erg cm^{-2} s^{-1})	Index	Notes
GRB 081213A						
	10^2	1	< 11.1	< 8.27	-2[a]	No detection (GTI $\sim 10^2$ s)
	10^3	0	< 3.69	< 2.75	-2[a]	No detection (GTI $\sim$281 s)
	10^4	0	< 0.548	< 0.408	-2[a]	No detection (GTI $\sim$2.84 ks)
GRB 180128A						
	10^2	…	…	…	…	Exposure = 0
	10^3	0	< 1.13	< 0.85	-2[a]	No detection; first exposure at $\sim$148 s
	10^4	0	< 0.315	< 0.234	-2[a]	No detection (GTI $\sim$5.52 ks; first exposure at $\sim$148 s)
GRB 200415A						
	10^2	10	< 12.6	< 9.42	-2[a]	Mild excess
	10^3	**25**	3.60 ± 2.00	4.14 ± 2.33	-1.71 ± 0.34	LAT detection ($\sim 5\sigma$); hard spectrum
	10^4	22	1.62 ± 0.96	1.97 ± 1.16	-1.67 ± 0.36	Detection (GTI $\sim$4.17 ks)
GRB 231115A						
	10^2	3	< 9.28	< 6.92	-2[a]	No detection (low TS)
	10^3	1	< 1.40	< 1.05	-2[a]	No detection
	10^4	1	< 0.95	< 0.707	-2[a]	No detection (exposure $\sim$6.1 ks)
GRB 120616A						
	10^2	…	…	…	…	ROI outside LAT FoV; exposure = 0
	10^3	…	…	…	…	ROI outside LAT FoV; exposure = 0
	10^4	1	< 0.67	< 0.50	-2[a]	No detection (exposure $\sim$4.2 ks; first exposure at $\sim$1447 s)
GRB 200423A						
	10^2	…	…	…	…	ROI outside LAT FoV; exposure = 0
	10^3	1	< 2.67	< 1.99	-2[a]	No detection; first exposure at $\sim$619 s
	10^4	0	< 0.34	< 0.25	-2[a]	No detection; first exposure at $\sim$619 s
GRB 231024A						
	10^2	0	< 73.1	< 54.5	-2[a]	No detection (no events in 10^2 s ROI)
	10^3	0	< 1.74	< 1.29	-2[a]	No detection (GTI $\sim 10^3$ s)
	10^4	0	< 0.37	< 0.28	-2[a]	No detection

Notes. Fluxes refer to 0.1–10 GeV.
[a] ULs are 95% CL profile-likelihood values with the photon index fixed to -2.

so that the probability of observing a triplet at least as compact as Δt is

$$F(\Delta t) = 1 - e^{-R\Delta t}\,(1 + R\Delta t) \tag{5}$$

(T.-P. Li & Y.-Q. Ma 1983; G. Principe et al. 2023; Y. Xing et al. 2024).

Following T.-P. Li & Y.-Q. Ma (1983), we quantify the statistical significance of each triplet using the Li and Ma test statistic, which accounts for both the observed photon excess and the estimated background rate in the ROI. The Li and Ma significance is expressed in Gaussian-equivalent standard deviations (σ). Separately, we report the analytic waiting-time false-alarm probability $p = F(\Delta t)$ of the triplet defined above. The statistical power of the test resides in the fact that a genuine GeV transient signal will have several photons within a short interval, yielding a Δt far smaller than expected from the sparse background rate at these positions (see Figure 3).

A unique aspect of our MGF sample is that four events (GRB 081213A, GRB 180128A, GRB 200415A, and GRB 231024A) originate from the same host galaxy, NGC 253, corresponding to multiple flares from a single underlying magnetar. Because these events share the same sky position, the standard triplet search, which examines photons from a fixed spatial region, can suffer from temporal cross contamination: photons from one flare may spuriously appear as appear as "delayed" triplets relative to an earlier trigger. In addition to the first-triplet search, we examine the most compact (minimum-Δt) triplet within a 1 yr post-trigger window. Because this is a minimum selected over the $\sim N_{\rm trip}$ triplets searched, it carries a look-elsewhere penalty, which we evaluate directly by a photon-time scramble: redistributing the photon arrival times uniformly over the livetime (corrected for bad time intervals) and recording the minimum triplet interval over 10^5 realizations yields a trials-corrected false-alarm probability p for each trigger window. The number of realizations is set a priori so that the Monte Carlo uncertainty on the significance is $<0.02\sigma$ even for the smallest scramble-derived p in the sample ($p = 0.005$ for GRB 231024A). The photon-time scramble directly provides these per-window trials-corrected probabilities; the analytic Erlang-2 probability $F(\Delta t)$ is reported separately as the single-triplet, pretrial compactness probability. Where noted, we further apply a Bonferroni correction over the six nondetected events.

3. Results

Table 2 summarizes the unbinned likelihood analysis results for each burst and integration window.

GRB 200415A. We detect a LAT source spatially consistent with the GBM localization in the longer integration windows. Across the $\Delta T = 10^2$–10^3 s windows, the TS values track the

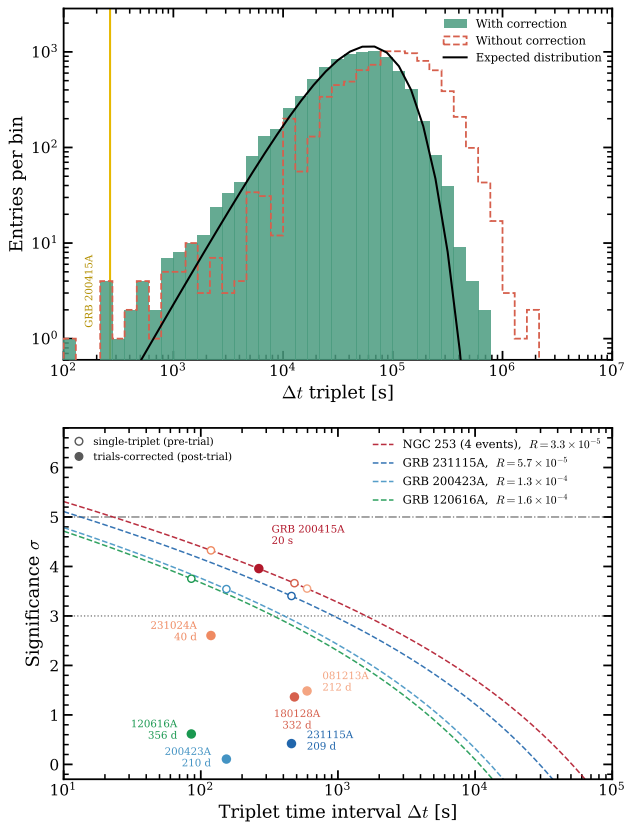


Figure 3. Photon-triplet analysis at the MGF positions. Top: distribution of the triplet time interval Δt for consecutive-photon triplets at NGC 253 (8375 triplets over 17 yr), with (green) and without (dashed) the correction for Fermi-LAT orbit and field-of-view gaps, compared to the distribution expected for independent Poisson events (solid black; Erlang-2). The correction shifts the distribution toward the expected form; the gold vertical line marks the GRB 200415A first post-trigger triplet ($\Delta t = 265$ s). Bottom: single-triplet waiting-time significance, $\sigma(\Delta t) = \Phi^{-1}[1 - F(\Delta t)]$ with F the Erlang-2 cumulative distribution function (CDF), as a function of Δt at each position's background rate R (dashed curves; the four NGC 253 events share one rate). Open circles mark the single-triplet (pretrial) value on each curve; filled circles mark the per-window trials-corrected (post-trial) value from the 1 yr photon-time scramble. The prespecified first triplet of GRB 200415A lies on its curve at 3.96σ, whereas the six next-best triplets fall below their curves and are at delays of 40–356 days, consistent with background.

onset of the emission, culminating in a detection with TS $\simeq$ 25 in the 10^3 s window. The measured photon flux is $F_\gamma = (3.6 \pm 2.0) \times 10^{-6}$ ph cm^{-2} s^{-1} with a hard photon index of $\Gamma = -1.71 \pm 0.34$. In the $\Delta T = 10^4$ s window, we continue to observe a significant excess (TS = 22), with $F_\gamma = (1.62 \pm 0.96) \times 10^{-6}$ ph cm^{-2} s^{-1} and $\Gamma = -1.67 \pm 0.36$. These properties are fully consistent with published LAT analyses of this event (H.-M. Zhang et al. 2020; V. Chand et al. 2021; Fermi-LAT Collaboration et al. 2021), which attribute the emission to synchrotron radiation from an external shock driven by relativistic ejecta or a shock resulting from collision with existing bow-shock shell.

Other candidates. We find no evidence of GeV emission from GRBs 231115A, 180128A, 120616A, 200423A, 231024A, or 081213A in any of the searched windows. Accordingly, we report ULs in Table 2. For three events, GRBs 180128A, 120616A, and 200423A, the earliest emission is unconstrained because during the shorter integration windows we have no exposure after standard selections. The first LAT exposure for these events begins at $\sim 1.5 \times 10^2$ s, $\sim 1.4 \times 10^3$ s, and $\sim 6.2 \times 10^2$ s, respectively. The limits reported here apply strictly to the observation times following the exposure. For these partial exposures, assuming no emission occurred during the unobserved interval, the reported limits are less constraining than those that would have been obtained from the full integration window.

Figure 1 displays the photon flux as a function of the integration time window δT for the full sample. Circles denote the detection of GRB 200415A, while inverted triangles represent ULs for the remaining candidates. The TS maps (generated with and without a point source at the host position) reveal no significant excesses beyond GRB 200415A at the target positions (Figure A1). The individual 500 s SEDs also yield only ULs for all candidates except GRB 200415A (Figure 2). To probe for emission below the individual detection thresholds, we considered the 500 s interval post-trigger and computed the stacked flux UL and the energy-resolved SEDs (in black in Figures 1–2).

The complete triplet analysis results are presented in Table 3. Among the seven candidates, only GRB 200415A shows significant prompt GeV emission: its first post-trigger triplet has a waiting-time false-alarm probability $p = 3.8 \times 10^{-5}$ (3.96σ), with a triplet time interval $\Delta t = 265$ s arriving within $\sim$20 s of the GBM trigger; the same triplet gives $\sigma = 5.39$ under the Li and Ma statistic evaluated for this triplet. This provides a complementary line of evidence for the likelihood-based detection (Fermi-LAT Collaboration et al. 2021).

The remaining six candidates show no significant triplet emission, with $\sigma < 1$ and $\Delta t > 10^4$ s in all cases (Table 3). Examining instead the most compact (minimum-Δt) triplet within a window from the trigger, the most significant is GRB 231024A ($\Delta t = 119$ s, 40.5 days post-trigger), with a per-window trials-corrected false-alarm probability $p = 0.005$ (2.6σ) from the photon-time scramble; after a Bonferroni correction over the six nondetection trigger windows, this becomes $p_{\rm global} = 0.03$ (1.9σ). All other windows reach at most 1.5σ, consistent with background (Figure 3). These results corroborate the likelihood-based nondetections.

4. Discussion

Phenomenological considerations. Our LAT analysis yields a single GeV detection—GRB 200415A—whose properties (Table 2) agree with previous work. The $\sim$20 s onset and emission extending over $\sim$20–400 s are consistent with synchrotron radiation from an external shock or bow shock driven by relativistic ejecta (H.-M. Zhang et al. 2020; V. Chand et al. 2021; Fermi-LAT Collaboration et al. 2021). For the remaining candidates, the nondetections are consistent across all four search methods (fixed-window likelihood, 500 s SED scans, stacking, and photon-triplet analysis), reinforcing the absence of GeV emission from these events. We note, however, that for three of the six nondetected candidates (GRBs 180128A, 120616A, and 200423A), the LAT exposure begins only at $\sim$148, $\sim$1447, and $\sim$619 s post-trigger, respectively.

In the 10^3 s interval, the 0.1–10 GeV energy-flux ULs are of order 10^{-9} erg cm^{-2} s^{-1}. In the 10^4 s window, the longer exposure yields deeper constraints: for GRB 120616A in IC 342 ($d \simeq 2.3$ Mpc) we obtain $F_E < 5 \times 10^{-10}$ erg cm^{-2} s^{-1}, a factor of $\sim$4 below the 10^4 s flux of GRB 200415A, while for GRBs 081213A and 231024A in NGC 253 ($d \simeq 3.7$ Mpc, the same host as GRB 200415A) we

Table 3
Photon-triplet Analysis Results

Source	Host	Next Triplet Analysis				Next-best Triplet Analysis		
		σ	p	Δt (s)	Delay (s)	Δt (s)	Delay (days)	p (σ)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
GRB 200415A	NGC 253	5.39	3.8×10^{-5}	265	20	265	∼0.0	3.8×10^{-5} (3.96)[a]
GRB 231115A	M82	0.81	0.85	81,340	1406	458	208.8	0.34 (0.4)
GRB 200423A	NGC 6946	0.58	0.34	16,153	7554	154	209.7	0.46 (0.1)
GRB 180128A	NGC 253	0.41	0.63	114,160	35,479	482	332.2	0.086 (1.4)
GRB 231024A	NGC 253	0.22	0.39	103,068	115,840	119	40.5	0.005 (2.6)[b]
GRB 081213A	NGC 253	0.14	0.33	84,045	63,965	595	211.8	0.069 (1.5)
GRB 120616A	IC 342	0.11	0.80	20,557	13,262	85	355.8	0.27 (0.6)

Notes. Photon-triplet analysis using Fermi-LAT data (0.1–100 GeV) within a 1° ROI. *The next triplet analysis* (columns (3)–(6)): Li and Ma significance (σ) and the analytic waiting-time false-alarm probability ($p = F(\Delta t)$), which are distinct statistics of the same triplet, triplet time interval ($\Delta t = t_3 - t_1$), and delay from trigger to first photon for the first triplet following each trigger. *The next-best triplet analysis* (columns (7)–(9)): the most compact (minimum-Δt) triplet within each 1 yr post-trigger window, its delay from trigger, and its per-window trials-corrected false-alarm probability p (with the equivalent Gaussian significance σ in parentheses) from a 10^5 realization of photon-time scramble (Figure 3). Only GRB 200415A shows significant prompt GeV emission ($\sigma = 5.39$, $p = 3.8 \times 10^{-5}$, delay ∼20 s).
[a] The next-best triplet of GRB 200415A is its prespecified prompt triplet; the quoted $p = 3.8 \times 10^{-5}$ (3.96σ) is the analytic waiting-time value of corresponding to that triplet.
[b] GRB 231024A is the most significant next-best triplet ($p = 0.005$, 2.6σ); after a Bonferroni correction over the six nondetected events, its significance decreases to 1.9σ.

Table 4
Derived Fireball Parameters and Constraints on the Baryonic Outflow Mass

GRB	$L_{\gamma,47}$ (10^{47} erg s^{-1})	$E_{\rm LAT,iso}$ (10^{46} erg)	T_0 (keV)	η_*	$M_{b,22}^{\rm (BP)}$ (10^{22} g)	$M_{b,22}^{\rm (LAT)}$ (10^{22} g)	$E_{\rm k,iso}/E_{\rm k,iso}$(200415A)
GRB 081213A	0.08	< 0.45	148	75	< 2.1	< 65.6	0.03
GRB 180128A	0.04	⋯	122	62	< 3.6	⋯	0.04
GRB 200415A	0.71	0.68[a]	252	129	< 40.8	> 58.3[a]	1.00
GRB 231024A	0.06	< 0.21	135	69	< 3.0	< 33.9	0.04
GRB 120616A	0.05	⋯	127	65	< 1.3	⋯	0.02
GRB 200423A	2.66	⋯	351	179	< 17.6	⋯	0.60
GRB 231115A	0.12	< 0.16	161	82	< 5.2	< 21.6	0.08

Notes. $L_{\gamma,47}$ is the isotropic gamma-ray luminosity in the GBM band, in units of 10^{47} erg s^{-1}, obtained using $E_{\gamma,\rm iso} = L_{\gamma,\rm iso} T_{90}$. The LAT constraints ($E_{\rm LAT,iso}$, $M_{b,22}^{\rm (LAT)}$) use the 10^3 s integration window (see Section 4) and are quoted only for events with LAT exposure from the trigger time. No LAT constraint is listed for GRBs 180128A, 200423A, and 120616A, whose first LAT exposure begins only at ∼148, ∼619, and ∼1447 s, respectively; for these events a 10^3 s isotropic-energy budget would require assuming emission during the unobserved early interval, which is precisely the epoch left unconstrained. $E_{\rm LAT,iso}$ is the 0.1 –10 GeV isotropic energy, in units of 10^{46} erg. $M_{b,22}^{\rm (BP)}$ is the UL on the baryonic mass from the baryonic-poor requirement $\eta > \eta_*$, and $M_{b,22}^{\rm (LAT)}$ is the mass constraint from the LAT band, both in units of 10^{22} g, assuming $\xi_\gamma = 0.3$ and $\xi_L = 0.1$. $E_{\rm k,iso}/E_{\rm k,iso}$(200415A) is the fiducial baryonic-rich kinetic energy normalized to GRB 200415A; for $\xi_\gamma = 0.3$ the reference value is $E_{\rm k,iso}(200415{\rm A}) = 3.31 \times 10^{46}$ erg, and absolute values are obtained by multiplying the tabulated entries by this reference value.
[a] For GRB 200415A, $E_{\rm LAT,iso}$ and $M_{b,22}^{\rm (LAT)}$ refer to the measured LAT emission and the corresponding *lower* bound on the baryonic mass. Consistency with the baryonic-poor limit implies a larger effective GeV efficiency for this event, as discussed in Section 4.

find $F_E \lesssim 3$–4×10^{-10} erg cm^{-2} s^{-1}. Since these events share the same distance as GRB 200415A, the flux ULs translate directly into luminosity constraints, ruling out a GeV counterpart of comparable brightness. The stacked-likelihood analysis further strengthens this conclusion: combining the five candidates with LAT exposure in the 10^3 s window yields $\mathrm{TS_{stack}} \approx 0$ and a UL $\hat{F}_{95} \approx 2 \times 10^{-10}$ erg cm^{-2} s^{-1} on the population-averaged 0.1–10 GeV energy flux, implying that the UL on the average GeV emission across the population is more than an order of magnitude below that of GRB 200415A.

Physical interpretation. For each MGF in our sample, we use the observed prompt emission to estimate basic fireball parameters following E. Nakar et al. (2005) and K. Ioka et al. (2005). We take $E_{\gamma,\rm iso}$ from the prompt measurements reported in the literature and infer the characteristic prompt-spike luminosity as $L_{\gamma,\rm iso} \equiv E_{\gamma,\rm iso}/T_{90} = 10^{47} L_{\gamma,47}$ erg s^{-1} (Table 1). A fraction ξ_γ of the total luminosity is assumed to emerge in gamma rays, so that the total fireball luminosity is $L_0 = L_{\gamma,\rm iso}/\xi_\gamma \simeq 3 \times 10^{47}\, \xi_{\gamma,-0.5}^{-1}\, L_{\gamma,47}$ erg s^{-1}, where we write $\xi_\gamma = 0.3\, \xi_{\gamma,-0.5}$ as a fiducial normalization (so that $1/\xi_\gamma = 1/0.3 \approx 3.3$, rounded to 3 in the prefactor). The energy is initially released in a compact region of size $r_0 = 10^6\, r_{0,6}$ cm (i.e., $r_{0,6} \equiv r_0/10^6$ cm) near the magnetar surface, producing a pair-rich, optically thick fireball. The initial (radiation-dominated) temperature is $T_0 = (L_0/4\pi r_0^2 c\, a)^{1/4} \simeq 275\ \xi_{\gamma,-0.5}^{-1/4}\, L_{\gamma,47}^{1/4}\ r_{0,6}^{-1/2}$ keV, where $a = 7.56 \times 10^{-15}$ erg cm^{-3} K^{-4} is the radiation constant. For the MGFs considered here, T_0 lies in the few $\times 10^2$ keV range (Table 4), and the observed $E_{\rm p}$ values are of the order of $3kT_0$ (H.-M. Zhang et al. 2020). The low-energy spectral indices of five candidates ($\alpha \approx -0.6$ to $+0.0$) are harder than the slow-cooling synchrotron limit ($\alpha = -2/3$; R. D. Preece et al. 1998), consistent with a quasi-thermal photospheric origin. The exceptions are GRBs 200423A and 231024A, whose indices

($\alpha = -0.7$ and -0.8, respectively) fall below this limit, admitting a nonthermal origin for these two events.

As the fireball expands and accelerates, the Thomson optical depth decreases and the flow eventually becomes transparent. The critical Lorentz factor corresponding to the photospheric transition can be computed as $\eta_* = (L_0 \sigma_T / 4\pi m_p c^3 r_0)^{1/4} \simeq 140\ \xi_{\gamma,-0.5}^{-1/4} L_{\gamma,47}^{1/4} r_{0,6}^{-1/4}$, where σ_T is the Thomson cross section and m_p is the proton mass, and the values we derive for each event are reported in Table 4. This critical value separates two limiting regimes for the outflow composition, parameterized by the dimensionless entropy $\eta \equiv L_0/\dot{M}c^2$. In particular, in the baryonic-poor regime ($\eta > \eta_*$)[5] the photosphere lies in the acceleration phase and the ejecta kinetic energy is suppressed relative to the injected fireball energy. This implies that the fireball radiates away the majority of its internal energy before it can fully accelerate. The terminal Lorentz factor saturates at $\Gamma_{ej} \simeq \eta_*$, and the fraction of the total luminosity converted into bulk kinetic energy is suppressed by the factor η_*/η (K. Ioka et al. 2005). Under these conditions, the isotropic baryonic kinetic energy is therefore limited to $E_{b,iso} \simeq (\eta_*/\eta)E_0$, which makes it possible to impose a strict physical UL on the baryonic mass M_b carried by the ejecta. Since a higher baryonic load would transition the fireball into the baryonic-rich regime, the mass must satisfy $M_b < M_b^{(BP)} \equiv (E_{\gamma,iso}/\xi_\gamma \eta_* c^2)$. Using the prompt parameters from Table 1 and the derived η_* values in Table 4, we find that the mass limits are $M_{b,22}^{(BP)} \lesssim 1.3$–$3.9$ (in units of 10^{22} g) for the fainter flares and $\lesssim 40.8$ for the bright event GRB 200415A. The inferred baryonic-poor parameters depend on the assumed prompt radiative efficiency ξ_γ and initial fireball radius r_0 as $\eta_* \propto \xi_\gamma^{-1/4} r_0^{-1/4}$ and $M_b^{(BP)} \propto \xi_\gamma^{-3/4} r_0^{1/4}$. Varying r_0 from 10 to 15 km changes η_* and $M_b^{(BP)}$ by only approximately 10%. Relative to the fiducial value $\xi_\gamma = 0.3$, adopting $\xi_\gamma = 0.1$, 0.5, and 1 changes the mass upper limits by factors of 2.28, 0.68, and 0.41, respectively. Thus, over the representative parameter range considered here, the mass limits vary by at most a factor of approximately 2.3 relative to the fiducial values and remain of the same order of magnitude.

We now compare the physical baryonic-poor requirement ($\eta > \eta_*$) with the observational constraints set by LAT nondetections. The latter provide ULs on the 0.1–10 GeV radiated energy and therefore on the shock-powered energy budget, which we express as limits on M_b (and equivalently on $E_{k,iso}$) for assumed efficiencies (K. Ioka et al. 2005; E. Nakar et al. 2005). Assuming that a fraction ξ_L of the baryonic kinetic energy is radiated in the GeV band via the shock-powered component, the observed LAT energy (or UL) implies a corresponding bound on the ejecta mass, $M_b < M_b^{(LAT)} = E_{LAT,iso}/(\xi_L \eta_* c^2)$. We adopt a fiducial conversion efficiency of $\xi_L = 0.1$, which renders the LAT-derived limits conservative. We convert the measured 0.1–10 GeV energy fluxes (and ULs) into isotropic-equivalent energies using the host distances in Table 1. Adopting the same bow-shock environment as GRB 200415A ($f \approx 50$), the five quasi-thermal candidates yield $t_{coll} \sim 10$ s, $\Gamma_{sh} \sim 10$, $t_\theta \sim 500$ s, and $E_{syn,max} \sim 1$ GeV. The predicted emission from the bow-shock interaction is anticipated within the first ~1000 s, and we therefore use our 10^3 s integration window for the LAT-derived constraints on the baryonic load. When the LAT-derived limit is weaker than the baryonic-poor upper bound (i.e., $M_b^{(LAT)} \gg M_b^{(BP)}$), the nondetection is expected: the physically allowed ejecta are intrinsically below current LAT sensitivity, and the observations do not probe the baryonic-poor constraint.

For the detected event GRB 200415A, the observed GeV emission requires a minimum baryonic load to power the signal. We find $M_b^{(LAT)} \gtrsim 5.8 \times 10^{23}$ g (lower limit), which is comparable to the baryonic-poor UL $M_b^{(BP)} \lesssim 4.1 \times 10^{23}$ g. Within the systematic uncertainties of the efficiency parameters, the convergence of these two bounds implies that GRB 200415A occurred in a marginally baryonic-poor regime where $\eta \sim \eta_*$. This parameter regime allows the fireball to be clean enough to exhibit high-E_p prompt emission, yet sufficiently loaded to carry the kinetic energy required for a detectable GeV afterglow. A modest increase in the effective GeV efficiency (larger ξ_L) reconciles the apparent tension between $M_b^{(LAT)}$ and $M_b^{(BP)}$ while retaining $\eta \sim \eta_*$.

Conversely, for the nondetected events, the physical constraints diverge significantly. For a representative case like GRB 081213A, the baryonic-poor physics restricts the mass to $M_{b,22}^{(BP)} < 2.1$. However, the sensitivity of the LAT corresponds to a much higher observational threshold of $M_{b,22}^{(LAT)} < 65.6$. Since the physically allowed mass is far below the sensitivity threshold ($M_b^{(BP)} \ll M_b^{(LAT)}$), the nondetections are a natural prediction of the model. Even considering the deeper limits achieved via stacking analysis, the predicted signal from these clean outflows remains below the detection limit. This conclusion still holds if we apply moderate changes in the assumed spectral index used to derive the limits. Thus, the lack of GeV detections for the broader population is consistent with the intrinsic faintness of baryonic-poor outflows and will remain undetectable with the current sensitivity.

The two candidates whose spectral indices fall below the slow-cooling synchrotron limit—GRBs 200423A ($\alpha = -0.7$) and 231024A ($\alpha = -0.8$)—may instead produce their prompt emission via optically thin processes such as synchrotron radiation from internal shocks or magnetic reconnection above the photosphere. We note that the typical uncertainties on α for these faint, short events ($\sigma_\alpha \sim 0.2$–0.3; A. C. Trigg et al. 2024, 2026) place the slow-cooling boundary within 1σ of both central values, so a photospheric origin cannot be firmly excluded. Nevertheless, we consider the implications of a nonthermal interpretation. In this case, the photospheric scaling $E_p \propto E_{iso}^{1/4}$ does not apply, the baryonic-poor requirement $\eta > \eta_*$ is not enforced, and the outflow may reside in the baryonic-rich regime ($\eta < \eta_*$), where the ejecta reach $\Gamma_{ej} \simeq \eta$ before becoming transparent and the kinetic energy is not suppressed, yielding $E_{k,iso} \simeq E_{\gamma,iso}(1/\xi_\gamma - 1)$ (Table 4). Alternatively, if the prompt emission is powered by magnetic reconnection, a substantial fraction of the outflow energy is dissipated during the prompt phase, leaving a reduced energy budget for any subsequent afterglow. GRB 200423A is the more energetic of the two ($E_{\gamma,iso} = 8.5 \times 10^{45}$ erg); however, the LAT field of view did not include this source until ~619 s post-trigger, precluding meaningful constraints on early GeV emission. GRB 231024A ($E_{\gamma,iso} = 5.5 \times 10^{44}$ erg) had LAT exposure from ~100 s, yet yields TS = 0 at all time windows (Table 2), consistent with either a reconnection-dominated prompt phase in which the available energy has already been radiated, or an intrinsically faint afterglow reflecting its order-of-magnitude lower isotropic energy relative to GRB 200415A.

[5] This regime is also the one required to reproduce the prompt scaling $E_p \propto E_{iso}^{1/4}$ reported for magnetar giant flares (H.-M. Zhang et al. 2020; A. C. Trigg et al. 2026), which requires a photosphere located within the acceleration phase (H.-M. Zhang et al. 2020).

5. Conclusions

We carried out a systematic Fermi-LAT search for high-energy (0.1–10 GeV) emission from seven nearby extragalactic MGF candidates identified in Fermi–GBM data (A. C. Trigg et al. 2026). For each target, we performed unbinned likelihood analyses in post-trigger integration windows of 10^2, 10^3, and 10^4 s, constructed standardized 500 s data products including TS maps and energy-resolved SEDs, and applied both stacked-likelihood and model-independent photon-triplet searches.

Among the seven candidates, only GRB 200415A shows significant GeV emission, with TS $\simeq$ 25 in the 10^3 s window and a hard spectrum ($\Gamma \simeq -1.7$) consistent with previous LAT analyses and with synchrotron radiation from an external shock or bow-shock interaction (H.-M. Zhang et al. 2020; V. Chand et al. 2021; Fermi-LAT Collaboration et al. 2021). The photon-triplet analysis provides a complementary line of evidence for this detection, with the first triplet arriving $\sim$20 s after the GBM trigger. For the remaining six candidates, no evidence of GeV emission is found in any searched interval or analysis method, with all triplet significances $\sigma < 1$, corroborating the likelihood-based nondetections. For individual events, we obtain 95% CL energy-flux ULs of order $F_E \sim 10^{-9}$ erg cm^{-2} s^{-1} assuming a fiducial $\Gamma = -2$ spectrum. Stacking the four nondetected events with LAT exposure in the 500 s interval, spanning distances $d \simeq$ 2.3–7.7 Mpc, yields a population-averaged UL of $\approx 2 \times 10^{-10}$ erg cm^{-2} s^{-1}, more than an order of magnitude below the measured flux of GRB 200415A. We note that for three events (GRBs 180128A, 120616A, and 200423A), the source position was outside the LAT field of view at trigger time; the reported limits apply only after the first LAT exposure and any earlier GeV emission remains unconstrained.

Within the relativistic fireball framework, the hard prompt spectra of five candidates favor baryonic-poor outflows ($\eta > \eta_*$), which restrict the ejecta mass to $M_b \lesssim 1$–4×10^{22} g. This stringent baryon limit suppresses the kinetic energy available to drive a GeV afterglow well below current LAT sensitivity, naturally explaining the nondetections even with the deeper constraints achieved through stacking. GRB 200415A is consistent with an outflow near the critical transition $\eta \sim \eta_*$, where the fireball is sufficiently clean to produce a high-E_p prompt spike yet carries enough baryonic load to power the observed GeV signal. Two candidates (GRBs 200423A and 231024A) have softer spectral indices, implying a nonthermal prompt origin and potentially a baryonic-rich regime; however, their nondetections are explained by either late LAT exposure onset or intrinsically low isotropic energy.

The results of this analysis suggest that, within the limited statistics of the present sample (one detection out of seven candidates), detectable GeV afterglows from MGFs may be restricted to the most energetic events with favorable observing conditions, and that the lack of GeV counterparts in the broader population is a natural prediction of the baryonic-poor fireball model rather than an observational limitation alone. However, for single events the number of GeV photons is very limited and the detection of such emission remains challenging with the current effective area of the LAT. As the extragalactic MGF sample continues to grow through ongoing GBM searches, future LAT observations and next-generation gamma-ray instruments with larger effective areas may better address whether GRB 200415A represents a rare extreme or the bright end of a broader population of GeV-emitting magnetar flares.

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Appendix
Supplementary Figure

Figure A1 shows the LAT TS maps (Section 2) for the six MGF candidates with LAT coverage in the post-trigger window; GRB 120616A is not shown because its first LAT exposure begins at $\sim$1447 s. Only GRB 200415A shows a significant excess at the host position, which the fitted point source accounts for; the other five candidates show no significant excess (Section 3).

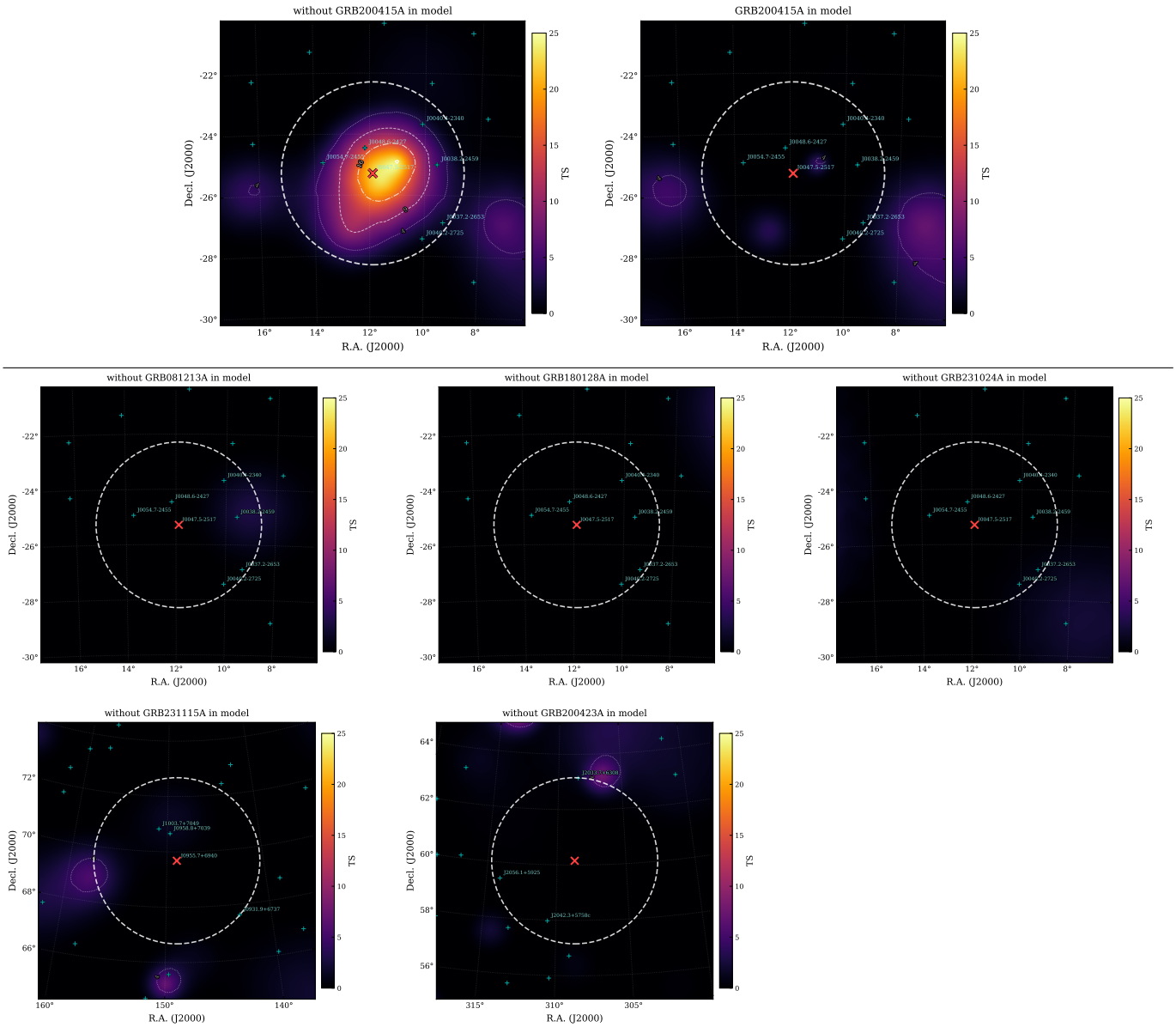


Figure A1. LAT TS maps in the post-trigger window for each MGF candidate. Top: the single detection, GRB 200415A, shown both without (left) and with (right) a transient point source at the host position; the with-source map illustrates that the fitted source accounts for the excess. Middle and bottom: the five nondetections (GRB 081213A, GRB 180128A, GRB 231024A, GRB 231115A, GRB 200423A), for which only the without-source map is shown, indicating the significance any excess would have to reach. Contours indicate TS levels of 4, 9, 16, and 25; the 3°-radius circle is centered on the host coordinates. The window is T_0 to $T_0 + 500$ s for all events except GRB 200423A, whose first LAT exposure begins at ∼619 s and which therefore uses $T_0 + 619$ to $T_0 + 1000$ s.

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